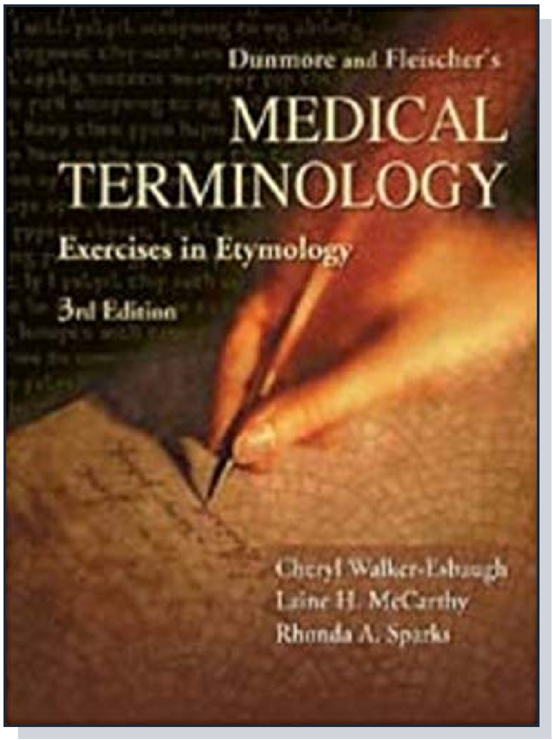
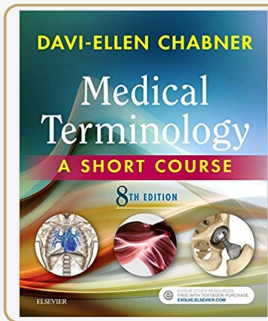
(eBook PDF) Dunmore and Fleischer's Medical Terminology Exercise in Etymology 3rd Edition



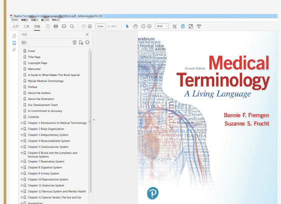
Home - Digital Library - Related Products

Related Study Materials



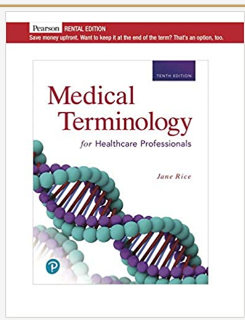
(eBook PDF) Medical Terminology A Short Course 8th Edition

Press this button to shop=>https://ebookluna.com/product/ebook-pdf-medical-terminolog



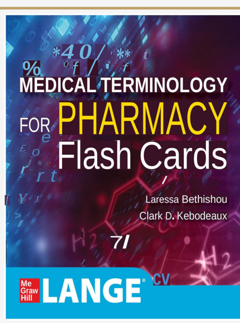
(Original PDF) Medical Terminology A Living Language (7th Edition)

Click to browse the store=>https://ebookluna.com/product/original-pdf-medical-termino



(eBook PDF) Medical Terminology for Healthcare Professionals, 10th edition

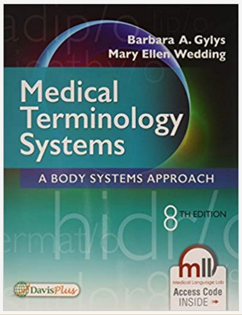
Click here for the storefront=>https://ebookluna.com/product/ebook-pdf-medical-termin



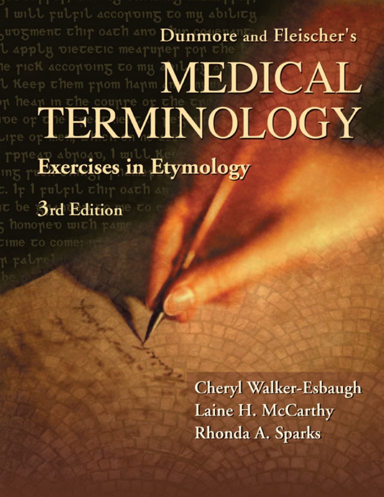
Medical Terminology for Pharmacy Flash Cards 1st Edition - eBook PDF

Tap to enter the online shop=>https://ebookluna.com/download/lange-medical-terminolog

Related Study Materials



(eBook PDF) Medical Terminology Systems: A Body Systems Approach 8th Edition



Dunmore and Fleischer's

MEDICAL TERMINOLOGY

Exercises in Etymology

3rd Edition

Cheryl Walker-Esbaugh
Laine H. McCarthy
Rhonda A. Sparks

Preface

In 1977, Charles W. Dunmore, an associate professor of classics at New York University, and Rita M. Fleischer, from the Latin/Greek Institute at City University of New York, published a novel approach to teaching the challenging language of medicine. Indeed, medicine does have a language all its own, based largely on a vocabulary drawn from ancient Greek and, to a lesser degree, Latin. This approach involved teaching students to recognize the roots of *medical terminology*, the etymology of the words health-care professionals use to communicate with each other and with patients. By teaching students the root elements of medical terminology—the prefixes, suffixes, and combining forms from Greek and Latin—Dunmore and Fleischer sought not only to teach students modern medical terminology but to give them the ability to decipher the evolving language of medicine throughout their careers.

In the third edition of this book, we have continued what Dunmore and Fleischer began. This new edition is organized essentially as Dunmore and Fleischer created it, with some important modifications to make it even more user friendly. The text is organized into interrelated units. Unit 1 (Lessons 1 through 7) includes 7 lessons based on Greek. Unit 2 (Lessons 8 and 9) is composed of 2 lessons based on Latin. Unit 3 (Lessons 10 through 15) takes a body systems approach that combines Greek and Latin elements used to describe the digestive system, respiratory system, and so forth. These first 15 lessons comprise the main body of the text. Each lesson builds and expands on the grammar and vocabulary introduced in the previous lessons.

For students who want additional exposure to medical terminology from a body systems perspective, the 4 lessons in Unit 4 provide just such an opportunity. These lessons include the hematopoietic and lymphatic, musculoskeletal, nervous, and endocrine systems.

Unit 5 stands on its own and provides an overview of *biological nomenclature*, the language used by scientists and physicians to identify the living organisms that exist in our world.

The pronunciation of medical terms follows the same rules that govern the pronunciation of all English words. The consonants *c* and *g* are "soft" before the vowels *e*, *i*, and *y*. That is, they are pronounced like the *c* and *g* of the words *cement* and *ginger*. Before *a*, *o*, and *u*, the consonants are "hard," and are pronounced like the *c* and *g* of *cardiac* and *gas*. The consonant *k* is always "hard," as in *leukocyte*. The long vowels *eta* and *omega* of Greek words are marked with the macrons ē and ō; this indicates that they are pronounced like the *e* and *o* of *hematoma*. Long *i* is pronounced "eye," as in the *-itis* of *appendicitis*. Words are pronounced with a stronger accent (emphasis) on one syllable. The accent falls on the second to last syllable if that syllable is long. To be considered long, a syllable must contain a short vowel followed by two consonants, a diphthong, or a long vowel (*neph-rī′-tis*). If the second to last syllable is short, the accent falls upon the third syllable from the end of the word (*gen′-ĕ-sis*).

The appendices have been expanded in this edition and include indexes of Latin and Greek suffixes, prefixes, and combining forms, as well as an abbreviated English-to-Greek/Latin glossary and a complete list of terms found in the exercises in Lessons 1 through 15. These appendices provide additional support for students and instructors alike.

The structure of the exercises at the end of each lesson has also changed from previous editions. All 15 of the major lessons contain 3 exercises. The first exercise asks students to analyze 50 terms based on the vocabulary found in that lesson. The second exercise requires students to identify a term based on its definition. The third exercise is a drill-and-review exercise and includes elements from the current and previous lessons.

This approach allows for smooth continuity and ensures that the major body of the text (Lessons 1 through 15) can be covered during a 1-semester course. The exercises in lessons 16 through 19 (Unit 4) are abbreviated and, for the most part, reflect only the material from that specific lesson. This approach allows these lessons to stand alone as additional study material. Lesson 20 (Unit 5), Biological Nomenclature, has also been written as a stand-alone lesson.

Terms in the lessons and exercises have been checked for currency and accuracy and verified in *Taber's Cyclopedic Medical Dictionary*, 19th edition (F.A. Davis Company, Philadelphia, 2001).

All 20 lessons include etymological notes to give students a historical perspective for medical terminology. These notes include tales from ancient Greek and Latin writers, mythical stories of gods and goddesses, excerpts from the writings of famous ancient physicians, such as Hippocrates and Celsus, and more modern stories of scientists and physicians who struggled to identify and accurately label the phenomena they observed.

This text is a workbook. We encourage you to write in this workbook and to make notes and comments that will help you as you work through the lessons and exercises.

Please note that this is not a medical textbook and should not be used for the diagnosis, treatment, prognosis, or etiology of disease. The medical content of this text, although accurate, is incomplete and is included solely to provide students with a context within which to learn medical terminology.

We hope you enjoy using this text to learn the complex and elegant language of medicine and that the knowledge you gain will benefit you throughout your career.

Cheryl Walker-Esbaugh
Laine H. McCarthy
Rhonda A. Sparks

Acknowledgments

We greatly appreciate the comments and suggestions from a number of people who gave their time and expertise. Special thanks to Drs. John Catlin and Ralph Doty, who, having taught with the second edition of the book, provided valuable insight into its revision and were always available to answer questions; and to Dr. Samuel Huskey, who read several of the revised chapters. Thanks to Dr. Dave McCarthy for reading and commenting on the biological nomenclature lesson and to Danny McMurphy, who patiently read chapter after chapter. Thanks, too, to Tracy Alford for her support and consideration.

We would like to thank F.A. Davis for their generosity in allowing us to use material in *Taber's Cyclopedic Medical Dictionary*, 19th edition. We also thank the editors at Davis for their advice and support. All of the people involved helped to make this a much better work than we could have ourselves. Any errors, of course, are our own.

Contents

Development of the English Language xiii

Unit 1 **Greek-Derived Medical Terminology** **1**
- Lesson 1 Greek Nouns and Adjectives 3
- Lesson 2 Nouns of the Third Declension 17
- Lesson 3 Building Greek Vocabulary I: Nouns and Adjectives 29
- Lesson 4 Greek Verbs 39
- Lesson 5 Building Greek Vocabulary II 51
- Lesson 6 Building Greek Vocabulary III 63
- Lesson 7 Building Greek Vocabulary IV 75

Unit 2 **Latin-Derived Medical Terminology** **85**
- Lesson 8 Latin Nouns and Adjectives 87
- Lesson 9 Latin Verbs 107

Unit 3 **Body Systems** **121**
- Lesson 10 Cardiovascular System 123
- Lesson 11 Respiratory System 137
- Lesson 12 Digestive System 149
- Lesson 13 Optic System 163
- Lesson 14 Female Reproductive System 175
- Lesson 15 Genitourinary System 187

Unit 4 **Additional Study** **201**
- Lesson 16 Hematopoietic and Lymphatic Systems 203
- Lesson 17 Musculoskeletal System 215
- Lesson 18 Nervous System 225
- Lesson 19 Endocrine System 237

Unit 5 **Biological Nomenclature** **245**
- Lesson 20 Biological Nomenclature 247

Appendices **255**
- Appendix A Index of Combining Forms 257
- Appendix B Index of Prefixes 269
- Appendix C Index of Suffixes 271
- Appendix D Index of Suffix Forms and Compound Suffix Forms 273
- Appendix E Glossary of English-to-Greek/Latin 275
- Appendix F Medical Terminology Used in Lessons 1 to 15 285

Bibliography for Edition III **299**

Bibliography for Edition II **301**

DEVELOPMENT OF THE ENGLISH LANGUAGE

In 55 and 54 BC, Julius Caesar invaded Britain. The romanization of Britain, however, did not occur until almost 100 years later, when expeditionary forces were sent out by the Roman emperor Claudius. Although Latin was the official language during the Roman occupation of Britain, Celtic, the native language of the people of Britain, was little affected by it.

The English language began its development as an independent tongue with the migration of Germanic people (Angles, Saxons, and Jutes) from western Europe (modern-day Denmark and northern Germany) across the English Channel to Britain during the 5th and 6th centuries AD. These Germanic invaders, in contact with the Romans from the 1st century BC on, brought with them not only their native tongue but also the Latin words they had borrowed from the Romans. Their language, known as Old English or Anglo-Saxon, was a member of the Germanic family of Indo-European languages and gradually superseded the Celtic dialects in most of southern Britain. Many Old English words have survived, with some linguistic change, to form the basic vocabulary of the English language. Words borrowed from other languages—mostly Latin, French, and Greek—have been added to the English language.

During the 7th century AD, after the establishment of the first monastery in 597 AD, the inhabitants of Britain gradually converted to Christianity. Latin, the language of the Western Church, was spoken, written, and read in the churches, schools, and monasteries. This brought many Latin words into the evolving English language, most having to do with religious matters and many derived from Greek.

Beginning in the 8th century AD, as a result of the Viking invasions, additional words of North Germanic origin entered the English language. Living alongside the Anglo-Saxons and eventually assimilated by them, the Norse and Danish invaders and their language had a marked impact on both England and the English language. It is not surprising that, in 1697 AD, writer Daniel Defoe described English as "your Roman-Saxon-Danish-Norman-English."

The Norman invasion in 1066 AD brought a French-speaking aristocracy to England, and for the next 150 years French was the official spoken and written language of the governing class. In this period, French, with its roots in Latin, existed alongside English but had little effect on it. However, in the 300 years following the expulsion of the Normans from England—from about 1200 to 1500 AD—although English was once again the dominant language, many words were borrowed from French because its vocabulary was far richer. Writers and educated people in England began to look to French as a source of words and concepts lacking in their own language. During these years, the changing English language reached the stage we now know as Middle English.

With the Renaissance (1400–1600 AD) came a revival of classic scholarship. English words began to be formed directly from Latin and Greek and were no longer borrowed through the intermediary of French. Beginning around 1500 AD, for the first time the writings of the ancient Greeks were read in England in their original language. This renewal of interest in Greek and Roman literature and the ideas and concepts expressed in these works created an awareness of the impoverished state of the English language and the difficulty of expressing in English ideas that were easily expressed in Latin or Greek. Words were borrowed extensively from Greek and Latin, both with and without change, and new words were created that combined both Latin and Greek elements. Although most words were borrowed from Latin and Greek, others were borrowed from French and Italian. The English of this period is now known as Modern English.

The extensive borrowing of words from Latin and Greek that began about 1500 AD continued for hundreds of years, and continues to this day. As new advances were made in the fields of medicine and science during and after the Renaissance (and continuing up to the present day), words were needed to describe these new discoveries and inventions. Medical scientists turned to the early Greek and Roman physicians, especially Hippocrates, Galen, and Celsus, whom they greatly admired, and borrowed words from their medical treatises. These ancient scientists had an extensive medical vocabulary, which they used to describe their observations and theories. Hippocrates, for example, used the terms *apoplexy*, *hypochondria*, *dysentery*, *ophthalmia*, *epilepsy*, and *asthma* to describe certain physical features and conditions that he observed. Modern physicians and scientists, who could not find an appropriate word to describe diseases that were unknown to early physicians, turned to the vocabulary of the ancient languages and created suitable terms.

The language used by Linnaeus, as well as by other scientists and scholars of the Renaissance and the period following, that is, the period after 1500 AD, is called New Latin. Scientists and writers, schooled in Classical Latin, tried to emulate the classical writers and to revive the style of Cicero and others. The term New Latin refers to words that have been created in the form of, and on the analogy of, Latin words (e.g., *natrium*, the chemical name for sodium, borrowed from Arabic) or to the use of new meanings applied to extant Latin words (e.g., the word cancer, from *cancer*; crab, or the word bacillus, from *bacillus*, a small rod or staff). New Latin is a rich source of biological terms. *Trichinella spiralis*, the species of *Trichinella* that causes trichinosis, and *Salmonella*, the genus of microorganisms named after the American pathologist Daniel E. Salmon, are two of the many examples of New Latin found in this text.

Figure 1. Apollo. (From Bulfinch's *Mythology—The Age of Fable,* with permission. Available at *http://www.bulfinch.org*).

HIPPOCRATES

Hippocrates, born in 460 BC, was a Greek physician who lived on the Aegean island of Cos. Although he is the most famous of the ancient physicians and is recognized as the "father of medicine," very little is actually known about him or his life. The *Hippocratic Corpus*, a work of about 60 medical treatises attributed to Hippocrates, most likely reflects the work of many physicians rather than that of Hippocrates alone. Hippocrates is recognized for separating superstition from medicine. Unlike other physicians of his time, he believed that illness had a rational explanation, rather than being the result of divine anger or possession by evil spirits, and could therefore be treated. Hippocrates based his medical writings on his observation and study of the human body. He was the first to attempt to record his experiences as a physician for future reference. The Hippocratic Oath, although it cannot be directly attributed to him, is said to reflect his philosophy and principles.

The Hippocratic Oath

"I swear by Apollo the physician, and Aesculapius, and Hygeia, and Panacea, and all the gods and goddesses, that according to my ability and judgment, I will keep this oath and its stipulation—to reckon him who taught me this art equally dear to me as my parents, to share my substance with him, and to relieve his necessities if required; to look upon his offspring in the same footing as my own brothers, and to teach them this art if they shall wish to learn it, without fee or stipulation, and that by precept, lecture, and every other mode of instruction, I will impart a knowledge of the art to my own sons, and those of my teachers, and to disciples bound by a stipulation and oath according to the law of medicine, but to none other.

"I will follow that system of regimen which, according to my ability and judgment, I consider for the benefit of my patients, and abstain from whatever is deleterious and mischievous. I will give no deadly medicine to anyone if asked, nor suggest any such counsel; and in like manner I will not give to a woman a pessary to produce abortion. With purity and with holiness I will pass my life and practice my art. I will not cut persons laboring under the stone, but will leave this to be done by men who are practitioners of this work. Into whatever houses I enter, I will go into them for the benefit of the sick and I will abstain from every voluntary act of mischief and corruption; and, further, from the seduction of females or males, of freemen and slaves. Whatever, in connection with my professional practice, or not in connection with it, I see or hear, in the life of men, which ought not to be spoken of abroad, I will not divulge, as reckoning that all such should be kept secret.

While I continue to keep this Oath unviolated, may it be granted to me to enjoy life and the practice of this art, respected by all men, in all times. But, should I trespass and violate this Oath, may the reverse be my lot." (From *Taber's Cyclopedic Medical Dictionary*, ed 19. FA Davis, Philadelphia, 2001, pp 949–950, with permission.)

GALEN

Galen (129–199 AD) was born in Pergamum in Asia Minor. After studying medicine at the Asclepium, the famed medical school in his native town, and in Smyrna and Alexandria, he came to Rome in 162 AD, where, except for brief interruptions, he remained until his death, writing philosophical treatises and medical books. His fame and reputation brought him to the attention of the emperor Marcus Aurelius, who appointed him court physician. Galen wrote extensively on anatomy, physiology, and general medicine, relying on his training, the best that was available, and on his dissection of human corpses and experiments on living animals. It was the work of Galen, more than any other medical writer, that profoundly influenced the physicians of the early Renaissance. His theories on the flow of blood in the human body remained unchallenged until the discovery of the circulation of the blood by William Harvey in the 17th century.

CELSUS

Aulus Cornelius Celsus was a Roman encyclopedist who, under the reign of the emperor Tiberius (14–37 AD), wrote a lengthy work dealing with agriculture, military tactics, medicine, rhetoric, and possibly philosophy and law. Apart from a few fragments, only his eight books on medicine still exist. It is suggested that Celsus was not a professional physician but rather a layman writing for other laymen. It appears, especially in his treatises on surgery, that he had little first-hand experience in the field of medicine and relied on material selected from other sources. Celsus was highly esteemed during the Renaissance, possibly as a result of his style of writing.

UNIT 1

GREEK-DERIVED MEDICAL TERMINOLOGY

GREEK NOUNS AND ADJECTIVES

NOUNS OF THE THIRD DECLENSION

BUILDING GREEK VOCABULARY I: NOUNS AND ADJECTIVES

GREEK VERBS

BUILDING GREEK VOCABULARY II

BUILDING GREEK VOCABULARY III

BUILDING GREEK VOCABULARY IV

Greek Alphabet

Name of Letter	Capital	Lower-case	Trans- literation	Name of Letter	Capital	Lower-case	Trans- literation
alpha	Α	α	a	xi	Ξ	ξ	x
beta	Β	ϐ or β	b	omicron	Ο	o	o short
gamma	Γ	γ	g	pi	Π	π	p
delta	Δ	δ	d	rho	Ρ	ρ	r
epsilon	Ε	ϵ	e short	sigma	Σ	σ or ς	s
zeta	Ζ	ζ	z	tau	Τ	τ	t
eta	Η	η	e long	upsilon	Υ	υ	y
theta	Θ	θ	th	phi	Φ	ϕ or φ	f, ph
iota	Ι	ι	i	chi	Χ	x	ch as in German "echt"
kappa	Κ	κ	k, c				
lambda	Λ	λ	l				
mu	Μ	μ	m	psi	Ψ	ψ	ps
nu	Ν	ν	n	omega	Ω	ω	o long

Source: Taber's Cyclopedic Medical Dictionary, ed 19, F. A. Davis, Philadelphia, 2001, p 2368, with permission.

LESSON 1

Greek Nouns and Adjectives

A man who is wise should consider health the most valuable of all things to mankind and learn how, by his own intelligence, to help himself in sickness.

[Hippocrates, *Regimen in Health* 9]

In the mid-8th century BC, the Greeks borrowed the art of writing from the Phoenicians, a Semitic-speaking people of the Levant who inhabited the region in the area of modern Lebanon. The Phoenician system of writing had to be adapted to the Greek language because there were characters representing sounds in the Semitic language that did not exist in Greek, and sounds in Greek for which there were no characters in the Semitic system. In their adaptation of these Phoenician characters, the Greeks began to distinguish between long and short vowels, representing long e by *eta* [H] and short e by *epsilon* [E], and long and short o by *omega* [W] and *omicron* [O], respectively. However, the distinction was carried no further, and no differentiation in writing was made between the long and short vowels a, i, and u. In transliterating Greek words, a macron (¯) will be used to mark the long vowels *ē* and *ō* in this text:

Greek	Meaning	Example
xēros	dry	**xeroderma**
splēn	spleen	**splenomegaly**
phōnē	voice	**phonology**
thōrax	chest cavity	**thoracentesis**

During and after the 1st century BC, many Greek words were borrowed by the Romans and, in the process of being borrowed, assumed the spelling of Latin words. It has been the practice since then, in the coining of English words from Greek, to use the form and spelling of Latin, even if the word never actually appeared in the Latin language.

The letter *k* was little used in Latin, and Greek *kappa* was transliterated in that language as *c*, which always had the hard sound of *k*. Most English words derived from Greek words containing a *kappa* are spelled with *c*:

Greek	Meaning	Example
kyanos	blue	**cyanotic**
mikros	small	**microscope**
kolon	colon	**colitis**
skleros	hard	**arteriosclerosis**

There are exceptions, and the *kappa* is retained as *k* in some words:

Greek	Meaning	Example
leukos	white	**leukemia**
kinēsis	motion	**dyskinesia**
karyon	kernel	**karyogenesis**
kēlē	swelling	**keloid**

Some words are spelled with either *k* or *c*: **keratocele, ceratocele** (*kerat*-, horn); **synkinesis, syncinesis** (*kin*-, move); cinematics, kinematics (*kinēma*, motion).

In Latin the letter *k* is rarely used and is found only in the following:

- *Kalendae*, the Calends, the first day of the month, and its derivatives *Kalendalis*, *Kalendaris*, *Kalendarium*, and *Kalendarius*
- *Karthago*, Carthage, the Phoenician city in North Africa
- *kalo* (archaic), call
- *koppa* (archaic), Greek symbol for 90

Greek words beginning with *rho* [*r*] were always accompanied by a strong expulsion of breath called **rough breathing** (also called aspiration). In transliterating Greek words and in the formation of English derivatives, this rough breathing is indicated by an *h* after the *r*:

Greek	Meaning	Example
rhombos	rhombus	**rhombencephalon**
rhodon	rose	**rhodopsin**
rhiza	root	**rhizoid**
rhythmos	rhythm	**rhythmic**

There are exceptions: **rhachis, rachis; rhachischisis, rachischisis** (*rhachis*, spine). Words beginning with *rho* [*r*] usually double the *r* when following a prefix or another word element, and the **rough breathing** [*h*] follows the second *r* (note: the following words are from Greek verbs):

Greek	Meaning	Example
rhe-	flow	**diarrhea**
rhag-	burst forth	**hemorrhage**
rhaph-	sew	**cystorrhaphy**

When Greek words containing diphthongs were borrowed and used in Latin, the diphthongs *ai*, *ei*, *oi*, and *ou* were changed to the Latin spelling of these sounds, but these Latin diphthongs usually undergo a further change in English:

Greek	Latin	English	Greek Example	Meaning	English Example
ai	*ae*	***e***	*haima*	blood	**hematology***
			aitia	cause	**etiology***
ei	*ei*	***ei*** or ***i***	*cheir*	hand	**cheirospasm, dyschiria**
			leios	smooth	**leiomyofibroma**
			meiōn	less	**miotic**
oi	*oe*	***e***	*oidēma*	swelling	**edema***
			oistros	desire	**estrogen**
ou	*u*	***u***	*gloutos*	buttock	**gluteal**

*British spelling usually retains the Latin diphthongs *ae* and *oe*: haematology, aetiology, oedema, oestrogen, and so forth.

In the Greek language, words beginning with the sound of the rough breathing [*h*] often lost their aspiration when another word element preceded the aspirated word (except after the prefixes *anti-*, *apo-*, *epi-*, *hypo-*, *kata-*, and *meta-*). However, the spelling of English derivatives of such words varies, and the aspiration is often retained:

Greek	Meaning	Example
hidros	sweat	**chromidrosis, hyperhidrosis**

Greek is an **inflected** language. This means that words have different endings to indicate their grammatical function in a sentence. The inflection of nouns, pronouns, and adjectives is called **declension**. Greek nouns are declined in five grammatical cases in both singular and plural: nominative, genitive, dative, accusative, and vocative. There are three declensions of Greek nouns, each having its own set of endings for the cases. Nouns are cited in dictionaries and vocabularies in the form of the nominative singular, often called the **dictionary form**.

Nouns of the first declension, mostly feminine, end in *-ē* or *-ā*, and sometimes in short *-a*. Second-declension nouns, mostly masculine or neuter, end in *-os* if masculine, and in *-on* if neuter. Third-declension nouns will be discussed in Lesson 2.

The base of nouns of the first and second declensions is found by dropping the ending of the nominative case, resulting in the **combining form**, to which suffixes and other combining forms are added to form words.

Greek	Meaning	Example
nephros	kidney	**nephr**-itis
neuron	nerve	**neur**-otic
psōra	sore	**psor**-iasis
psychē	mind	**psych**-osis

Rarely, the entire word is used as the combining form: **colonoscopy** (*kolon*, colon), **neuronitis** (*neuron*, nerve).

If a suffix or a combining form that begins with a consonant is attached to a combining form that ends in a consonant, then a vowel, called the **connecting vowel**, usually **o** and sometimes **i** or **u** (especially with words derived from Latin), is inserted between the two forms.*

leuk-**o**-cyte

neur-**o**-blast

psych-**o**-neurosis

calc-**i**-penia

vir-**u**-lent

Exceptions occur when suffixes beginning with *s* or *t* follow an element ending with *p* or *c*:

eclamp-sia

apoplec-tic

epilep-tic

nephremphraxis (for **nephremphrac-sis**)

Adjectives agree in gender, number, and case with the noun they modify. They are cited in dictionaries and

*Medical dictionaries and English dictionaries usually give the connecting vowel as part of the combining form, as in leuko-, neuro-, psycho-, and so forth.

vocabularies in the form of the nominative singular masculine. The dictionary form of most Greek adjectives ends in *-os*, and the combining form is found by dropping this ending. There are some adjectives that end in *-ys*, and the combining form of these is found by dropping the *-s* or, rarely, the *-ys*:

Greek	Meaning	Example
leukos	white	**leuk**-emia
kyanos	blue	**cyan**-osis
tachys	swift	**tachy**-pnea
glykys	sweet	**glyc**-emia

When Greek nouns are used in English, they usually appear in one of four ways:

1. In the original vocabulary form:
   kolon **colon**
   mania **mania**
   omphalos **omphalos**
   psychē **psyche**
2. With the ending changed to the Latin form:
   aortē **aorta**
   bronchos **bronchus**
   kranion **cranium**
   tetanos **tetanus**
3. With the ending changed to silent -e:
   gangraina **gangrene**
   kyklos **cycle**
   tonos **tone**
   zōnē **zone**
4. With the ending dropped:
   organon **organ**
   orgasmos **orgasm**
   spasmos **spasm**
   stomachos **stomach**

PREFIXES

Prefixes modify or qualify in some way the meaning of the word to which they are affixed. It is often difficult to assign a single specific meaning to each prefix, and often it is necessary to adapt a meaning that fits the particular use of a word. A complete list of prefixes is found in Appendix B.

a- (**an**- before a vowel or *h*): not, without, lacking, deficient:
a-biogenesis **an**-algesia
a-sthenia **an**-hydrous
cardi-**a**-sthenia **an**-hidrosis

anti- (**ant**- often before a vowel or *h*; hyphenated before *i*): against, opposed to, preventing, relieving:
anti-biotic **anti**-retroviral
anti-histamine **anti**-septic
anti-toxin **ant**-acid

di- (rarely **dis**-): two, twice, double:
di-phonia **di**-ataxia
di-plegia **dis**-diaclast

dys-: difficult, painful, defective, abnormal:
dys-menorrhea **dys**-pepsia
dys-pnea **dys**-genesis
dys-ostosis **dys**-trophy

ec- (**ex**- before a vowel): out of, away from:
ec-tasis **ex**-encephalia
ec-topic **ex**-ophthalmos

The prefix **ex-** in most words is derived from Latin: **excrete**, **exhale**, **extensor**, **exudate**, and so forth.

ecto- (**ect**- often before a vowel): outside of:
ecto-derm **ecto**-plasm
ecto-cornea **ect**-ostosis

en- (**em**- before *b*, *m*, and *p*): in, into, within:
en-cephalitis **em**-metropia
em-bolism **em**-physema

endo-, **ento**- (**end**-, **ent**- before a vowel): within:
endo-genous **ento**-cele
endo-metritis **end**-odontics
endo-cardium **ent**-optic

epi- (**ep**- before a vowel or *h*): upon, over, above:
epi-cardium **epi**-demic
epi-dermis **ep**-encephalon

exo-: outside, from the outside, toward the outside:
exo-cardia **exo**-genous
exo-crine **exo**-thermal

hemi-: half, partial; (often) one side of the body:
hemi-cardia **hemi**-paralysis
hemi-plegia **hemi**-gastrectomy

hyper-: over, above, excessive, beyond normal:
hyper-hidrosis **hyper**-lipemia
hyper-glycemia **hyper**-parathyroidism

hypo- (**hyp**- before a vowel or *h*): under, deficient, below normal:
hypo-chondria **hyp**-algesia
hypo-dermic **hyp**-acousia

mono- (**mon**- before a vowel or *h*): one, single:
mono-blast **mono**-chromatic
mon-ocular **mono**-neuritis

Another Random Scribd Document with Unrelated Content